Department of Electrical Engineering

Indian Institute of Technology, Madras

EE 2005 Electrical Machines - Quiz II

Class: II Year B.Tech
Duration: 50 min.

Date: 20th February 2024
Max. Marks: 20

Instructions to Candidates

1. All the answers to be written in the question paper itself.
2. For questions 1 –4, pick the most appropriate alternative(s) from among the options (A,B,C,D). More than one alternative may be correct and marks will be given only if all the correct option/ options (only) is/(are) written down at appropriate place.
3. All numerical computations must be done to 2 digits of precision.

Q5.-Q7 Write the steps clearly. Only final answer is not sufficient for the evaluation.

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1. What will be the current passing through a coil of 800 Turns wound around an iron ring generating Ampere Turns of 3200 A-T?

a) 4 A
b) 2 A
c) 1 A
d) 0.5 A

Ans: The correct option/(options) is/(are) ..... (a) ..... . **(Marks 1)**

2. Mr Amrit Singh purchased a transformer designed for 120 / 230 V, 60 Hz in the USA to operate a 230 V appliance purchased in India. However, on migrating back after some years, he attempts to use the same transformer in India as a 230 / 120 V transformer operated at 50Hz to power equipment meant for 120 V operation purchased in the USA. He is likely to find that the transformer

$\phi \propto V/f$

a) works properly - no problem.
b) draws increased current and gets heated up.
c) delivers voltage higher than 120 V.
d) is unable to draw current from mains supply.

Ans: The correct option/(options) is/(are) ..... (b) ..... . **(Marks 1)**

3. During the short circuit test on a small transformer the frequency is increased from 50Hz to 100Hz. The copper losses of the transformer

a) will increase by a factor of 4.
b) will increase by a factor of 8.
c) will increase by a factor of 1/4.
d) will almost remains the same.

Ans: The correct option/(options) is/(are) ..... (d) ..... **(Marks 1)**

4. Mr Dinakar Rao bought a 200/400 V, 50 Hz single phase transformer from Mr. Sadhu Seth. Mr Sadhu claimed that he designed the transformer such a way that the transformer has negligible core loss at 50Hz. Mr. Dinakar applied a 100 V supply at the primary side keeping 400 V side open. But to his astonishment he observed that the transformer was drawing 50Hz triangular shaped current from the primary side. This is because

$e = N d\phi/dt$

$\phi \propto \int e\,dt$

a) The supply voltage was sinusoidal wave at 50Hz frequency.
b) The supply voltage was triangular wave at 50Hz frequency.
c) The supply voltage was square wave at 50Hz frequency.
d) The supply voltage was saw tooth wave at 100Hz frequency.

Ans: The correct option/(options) is/(are) ..... (c) ..... . **(Marks 1)**

5. A steel ring 20 cm mean radius and circular cross section of 4 cm diameter has an air gap of 1 mm length. The core is uniformly wound with 500 Turns of wire carrying 10 A current. The air gap takes 60% of the total mmf.
   a. Find the reluctance of the steel ring.
   b. Find the relative permeability of the steel ring.

Leakage, fringing can be neglected.

$e = 5000$

$e_g = 3000 \quad e_s = 2000$

$A = 1.256 \times 10^{-3} m^2$

$3000 = \phi . \ell / \mu A \Rightarrow \phi = 4.74 \times 10^{-3}$ Wb

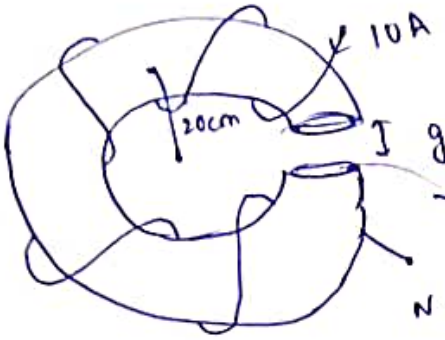
$R_S = \frac{e_s}{\phi} = 4.22 \times 10^5$ AT/Wb

a) **The Reluctance of the steel ring is** 4.221×10^7 **AT/Wb.** **(Marks 3)**

b) **The relative permeability of the steel ring is** 18.85. **(Marks 2)**

$\ell = 2\pi r = 1.26$

$R_S = \frac{\ell}{\mu_0 \mu_r A} \quad \mu_r = 1900$



→ mmf divides in the ratio of reluctances

$\Rightarrow \frac{R_{air\ gap}}{R_{total}} = 60\% = 0.6 \Rightarrow \frac{R_{air\ gap}}{R_{airgap} + R_{ring}} = 0.6$

$\Rightarrow 2 R_{air\ gap} = 3 R_{ring}$

total mmf $= Ni = 500 \times 10 = 5000$

$R_{airgap} = \frac{g}{\mu_0 A} = \frac{1 \times 10^{-3}}{4\pi \times 10^{-7} \times \pi \times 4 \times 10^{-4}} = \frac{10^8}{16\pi^2} = 6.332 \times 10^7$ At Wb^{-1}

$\Rightarrow R_{ring} = \frac{2}{3} \times 6.332 \times 10^7 = \boxed{4.221 \times 10^7 \text{ At Wb}^{-1}}$

$= \frac{\ell}{\mu A} = \frac{\pi \times 20 \times 10^{-2}}{\mu_r \times 4\pi \times 10^{-7} \times \pi \times 4 \times 10^{-4}}$

$= \frac{10^{10}}{\mu_r \times 4\pi}$

$\Rightarrow \mu_r = \frac{10^3}{4.221 \times 4\pi} = \boxed{18.85}$

$e = N d\phi/dt$ $\quad 200\sqrt{2}\sin(100\pi t) = 200\, d\phi/dt$

$\Rightarrow \phi = \frac{\sqrt{2}\cos(100\pi t)}{100\pi}$ $\quad \phi_{max} = 4.501\times10^{-3}$ Wb

6. A 200 Volt, 50 Hz transformer has 200 primary side turns. The transformer draws 5 A at 0.2 p.f. lagging current at no-load from the primary side when 200 Volt is applied at the primary side. $1 - 4.89j$ (i_c, i_m)

Determine: a) Maximum value of the flux in the core, b) Core loss, c) Magnetizing current, d) Core loss resistance, e) Magnetizing reactance. Both of them are seen from the 200 Volt side.

a) **Maximum value of the flux** ---- 4.501×10^{-3} ---- **Wb.** **(Marks 1)**

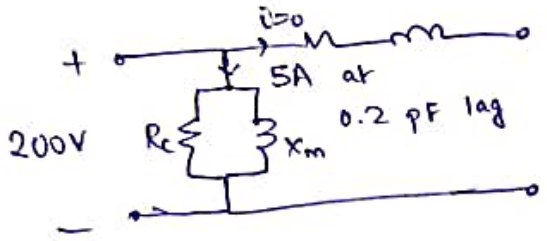
b) **Core loss** ---- 200 ---- **Watts.** **(Marks 1)** $V \cdot I_C$

c) **Magnetizing current** ---- 4.89 ---- **A.** **(Marks 1)**

d) **Core loss resistance** ---- 200 ---- **Ω.** V/I_C **(Marks 1)**

e) **Core loss reactance** ---- 40.89 ---- **Ω.** V/i_m **(Marks 1)**

$N_1 = 200$. At no-load, this is shunt current.



$i_{shunt} = 5\angle -\cos^{-1}(0.2) = 1 - 4.89j$ A

$= i_{core} + j \times i_m \Rightarrow \boxed{i_m = 4.89A}$

$\Rightarrow R_c = \frac{200}{1} = \boxed{200\ \Omega}$ $\quad \frac{200}{jX_m} = -4.89j$

$\Rightarrow X_m = \frac{200}{4.89} = \boxed{40.89\ \Omega}$

Core loss $= i_{core}^2 R_c = 1^2 \times 200 = \boxed{200\ W}$

If no load, $i_{secondary} = 0 \Rightarrow i_{primary} = 0$

$e = N\frac{d\phi}{dt}$

$\Rightarrow 200\sqrt{2}\sin(2\pi \times 50t) = 200 \cdot \frac{d\phi}{dt} \Rightarrow \phi = \int_0^t \sqrt{2}\sin(100\pi t)\, dt$

$\Rightarrow \boxed{\phi_{max} = 4.501 \times 10^{-3}\ Wb}$ $= \frac{\sqrt{2}\cos(100\pi t)}{100\pi} - 0$

7. A 5 kVA, 2500/500V, single phase transformer has following parameters.
HV Side: $R_{HV} = 10\Omega$, $X_{HV} = 20\ \Omega$.
LV Side: $R_{LV} = 0.2\Omega$, $X_{LV} = 0.5\Omega$.
The primary side voltage is kept at 2500 Volts during all the operating conditions.

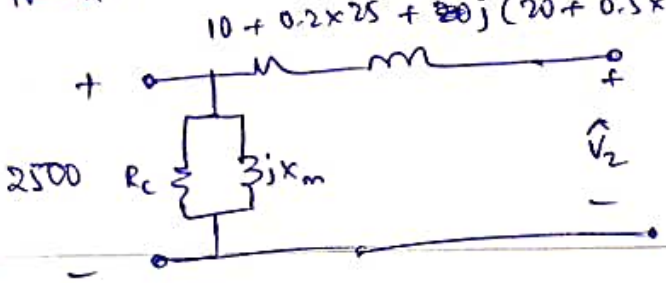
a. Find the **voltage regulation** in percentage and **secondary side terminal voltage** seen from the secondary side itself at full load at 0.8 p.f. lagging.
b. Find the **power factor** where zero voltage regulation will happen at full load.
c. Find the **efficiency** and **core loss resistance** of the transformer seen from the HV side when the voltage regulation is zero as discussed in problem 7(b). It is interesting to note that maximum efficiency of the transformer also happens in this operating condition.

a) **Voltage regulation** ---2.52---**%. Secondary side voltage** ---487.4---**Volts.** **(Marks 2)**

b) **Power factor**---0.908---. **(Marks2)**

c) **Efficiency** ---98.69---**%. Core loss resistance**---0---**Ω.** **(Marks 2)**

Refer to HV:



$$i_{full\ load} = \frac{5kVA}{2500} \quad (\text{on HV}) = \frac{5\times10^3}{2500} = 2A$$

$$VR = \frac{2500 - 2(15\times\cos\phi + 32.5\times\sin\phi)}{2500} \times 100\%$$

$$= \frac{2\times(15\times0.8 + 32.5\times0.6)}{2500} \times 100\% = \boxed{2.52\%}$$

$$\text{Sec}(V_2)_H = 2500 - 2\times(15\times0.8 + 32.5\times0.6) = \boxed{2437\ V}$$

$$V_{2L} \times \frac{500}{2500} = \boxed{487.4\ V}$$

If zero voltage regulation, $15\cos\phi \mp 32.5\sin\phi = 0$ (if $\cos\phi$ leading)

$$\Rightarrow \phi = 24.77 \Rightarrow PF = \cos(24.77) = 0.9079$$

Let's do it at HV side:

output power = $V_2 I_2 \cos\phi$

$= \frac{2500}{2437} \times 2 \times 0.908$

$= \boxed{4540\ W}$

Cu loss : $Re(V \cdot I^*) = Re(IR \cdot I^*) = |I|^2 \cdot Re(R)$

$= 2^2 \times (10 + 5) = \boxed{60\ W}$

Iron loss : When regulation is zero, there passes no current through $R_c \parallel jX_m$. (Z_ϕ, the shunt branch)

So loss = 0 W

$\Rightarrow \eta = \frac{P_{output}}{P_{input}} \times 100\ \%$

$= \frac{P_{output}}{P_{output} + losses} \times 100\ \%$

Core loss = Iron loss

$= \frac{4540}{4540 + 60 + 0} \times 100\ \%$

60

$= \boxed{98.69\ \%}$